The BR-PB-TV Decomposition, Its Identification, and Its Reductions

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Purpose. This white paper consolidates the BR-PB-TV component decomposition of the symptom score (documents 24 and 28), sets out how trial design identifies the components and why the analysis model need not mirror them, places the decomposition in the clinical trials and N-of-1 literatures, and reports the two two-component reductions of papers 13 and 14, with revised guidance on when each form is defensible.

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Notation index and glossary

Symbols follow `analysis/report/NOTATION.md` where it defines one (status *canonical*). Symbols local to this document are marked *local*. Code identifiers appear only in backticks.

Table 1: Notation index for this document, with meaning and status

Symbol	Meaning	Status
i, t	participant index; timepoint (weekly occasion) index	canonical
n	number of measurement occasions per participant	local
N	total number of participants across all randomization paths	canonical
Y_{it}	observed symptom score (code column `Sx`)	canonical
BL_i	participant baseline level	canonical
BR_{it}, PB_{it}, TV_{it}	biological, placebo-belief and natural-history response components, each a non-negative reduction in symptoms	canonical
ε_{it}	observation-level residual	canonical
u_{0i}, u_{1i}	participant random intercept and random time slope (section 23)	canonical (u_{0i} as u_i); local (u_{1i})
$D_{bc,it}$	exposure-decayed drug indicator: 1 on drug, $e^{-\lambda t_{sd}}$ off drug (code `Dbc`)	canonical
t_{sd}, t_{od}, t_{pb}	time since discontinuation; time on drug in the current on-drug period, reset to zero at re-exposure; time-by-belief accumulation	canonical
$t_{1/2}$, λ	carryover half-life; decay rate $\ln 2 / t_{1/2}$	canonical

Symbol	Meaning	Status
η, η_t, η(phase)	expectancy: the design expectancy at occasion t (1 open label, 0.5 blinded, 0 at baseline; code column `e`), written η(phase) where it is constant within a phase; it scales the PB mean and standard deviation	canonical
m, r, d	Gompertz maximum, onset rate and displacement (code `maxr`, `rate`, `disp`); subscripted m_{BR}, m_{PB}, m_{TV} etc. by component	local
$\sigma_{PB}^{\text{baseline}}$	PB standard deviation at full expectancy	local
β_t	slope of a linear-in-time straw-man response (section 2)	local
c_{cf1t}, c_{cfct}	cross-component correlation at the same and at different timepoints (code `c.cf1t`, `c.cfct`)	local
c_1, $c_\times$	the same two correlations as written in papers 13 and 14 (section 20)	local
ρ	within-component serial correlation (AR(1); the common correlation under compound symmetry in section 20)	canonical
c_{bm}	on-drug biomarker-BR correlation of the covariance-moderation architecture; c_{bm}^* its positive-definiteness ceiling	canonical
$c_{bm,PB}$	biomarker-PB correlation of paper 06's contamination studies (code `c.bm.pb`)	local (paper 06)

Symbol	Meaning	Status
M, v	component correlation block and unit-scaled biomarker coupling vector in the ceiling formula	local
β_{bm}^{BR}, β_{bm}^{PB}, β_{bm}^{TV}	component-specific biomarker slopes	canonical
$\beta_{bm}^{\text{lumped}}$	one-component (lumped) biomarker slope	canonical
w_{PB}, w_{TV}	design-dependent weights in the omitted-variable identity	local
B_i	participant baseline biomarker value (code `bm`)	canonical
R_i	participant total response in the illustrative regression of section 24	local
β_0; β_0^{BR}, β_0^{PB}, β_0^{TV}	intercepts (sections 23 and 24)	local
ξ_i, ξ_i^{BR}, ξ_i^{PB}, ξ_i^{TV}	regression errors in section 24	local
$\beta_{bm:D}$	biomarker-by-treatment interaction estimand (`bm:Dbc`); true value -2.25 in paper 06	canonical
$\lambda_{\min}(M)$	smallest eigenvalue of M (section 20); not the decay rate	local
α	nominal significance level, 0.05	canonical

Glossary. *BR*: biological (pharmacological) response. *PB*: placebo-belief (expectancy) response. *TV*: time-variant natural history. *Hybrid design*: open-label titration, blinded discontinuation, then a brief crossover (Hendrickson et al. 2020). *OL*, *CO*, *OL+BDC*: open-label, crossover, open-label plus blinded discontinuation. *Balanced-placebo (open-hidden) design*: a design that crosses drug exposure with told allocation, so belief varies independently of drug. *One-component (lumped) analysis*: the LME `Sx ~ bm + t + Dbc + bm:Dbc` with a random intercept and `corCAR1` residuals. *Phase-augmented analysis*: the

lumped model plus a phase factor and its interactions with `Dbc` and `bm:Dbc`. *Flattening*: setting a component's Gompertz maximum to zero while keeping its variance and covariance blocks. *Evidence labels*: *derived* (follows algebraically), *simulated* (Monte Carlo result from a cited driver), *inspected* (read in the source), *argued* (reasoned position).

0. What this document is

This white paper consolidates two earlier documents and extends them in two directions.

Document 24, the component decomposition pedagogy, supplies Parts I through III. It develops the three-component BR-PB-TV decomposition with worked examples and an intuition narrative. Document 24 remains in place as a standalone introduction and is not superseded. A reader meeting the decomposition for the first time may prefer it, since it is shorter and written to be read straight through. Parts I through III reproduce its content so that the present document is complete on its own.

Document 28, on the Hendrickson approach and the hybrid design, is folded into Parts II and III and is superseded.

Two parts are new. Part IV places the decomposition in the context of the clinical trials and N-of-1 literatures, and asks which of its three components are conventional and which are innovations of this program. Part V reports papers 13 and 14, which evaluate the two available two-component reductions.

We should state the organizing observation at the outset, since it runs through both new parts. The three components are not equally well established. A trend term is standard in the crossover and N-of-1 literature. A modeled placebo-belief trajectory is not, and the empirical placebo literature regards belief and natural history as separable only under designs that most trials do not field. That asymmetry turns out to govern which reductions of the decomposition are defensible.

Reading order

Table 2: Reading order of the six parts, with content and source documents

Part	Content	Source
I	The three components and why they are separated	doc 24
II	Identification from trial design	docs 24, 28

Part	Content	Source
III	Analysis models, and why they need not match the DGP	docs 24, 28
IV	**The decomposition in the wider literature**	**new**
V	**The two-component reductions**	**papers 13, 14 (new)**
VI	Reference: questions, summary table, bibliography	doc 24

A reader familiar with the decomposition may begin at Part IV. A reader who wants only the practical guidance may read section 21, which supersedes the earlier guidance of sections 12 and 13.

A note on maintenance

Parts I through III duplicate document 24, which is maintained separately. Document 24 is canonical for that material. A correction to the components pedagogy should be made there first and mirrored here.

Part I. The three components

1. Why split the response?

Imagine you are a PTSD patient trying prazosin (a blood-pressure medication that also reduces nightmares). After eight weeks of treatment, your symptom score has dropped by twelve points. The clinical question is straightforward: 'is the drug working?' The statistical question is harder: 'how much of that twelve-point drop should I attribute to prazosin, and how much to other things?'

Three plausible mechanisms produce a drop in symptom score on a clinical trial:

1. **Drug effect.** The medication chemically modulates the physiology of nightmares. Prazosin is an alpha-1-adrenergic antagonist that reduces noradrenergic tone during REM sleep, and its pharmacological action is independent of whether you know you are taking it.
2. **Expectation effect.** You believe the treatment will work, and your belief activates well-characterized neurobiological pathways: dopaminergic

reward signaling, descending pain modulation, anticipatory cortisol regulation. The placebo literature is now thirty years deep on this; it is not 'all in your head' in any dismissive sense.

3. **Natural history.** Some symptoms wax and wane on their own. PTSD severity drifts with anniversaries, life events, sleep hygiene, and the slow erosion that any chronic condition shows when measured repeatedly over months.

If we lump these three causes together as a single 'treatment response', we cannot distinguish them. The patient's twelve-point drop might be eight points of drug, three of expectation, and one of natural drift; or it might be two of drug, eight of expectation, and two of drift. The total is the same. The clinical implications are completely different.

The BR-PB-TV decomposition addresses this directly. It writes the observed change in outcome as a sum of three biologically interpretable components plus measurement noise:

$$Y_{it} \;=\; \mathrm{BL}_i - [\, BR_{it} + PB_{it} + TV_{it} \,] + \varepsilon_{it}.$$

Here BL_i is participant i's baseline symptom score, BR, PB, and TV are the three response components defined below (each a non-negative reduction in symptoms), and ε is observation-level noise. By decomposing the response into these three causes, we can ask cause-specific questions that the lumped form cannot answer: how much of the improvement is pharmacological, how much is expectation-driven, and how much would have occurred without any treatment at all.

The problem with one-component models

A naive analysis writes:

```
improvement = treatment_effect + error
```

The mathematical mistake is subtle but consequential. The 'error' term in such a model becomes a wastebasket for everything the analyst did not measure. If improvement is, in truth,

```
improvement = 4 (drug) + 6 (expectation) + 1 (natural history) + 1 (noise) = 12
```

the naive model fits this as

```
improvement = 12 (treatment_effect) + 0 (error)
```

at the population mean and infers that the drug is three times more powerful than it actually is. The structure is missed, and the inference is biased toward the lumped quantity rather than the quantity of clinical interest. Worse, the lumped quantity does not generalize: a population with stronger placebo expectations would appear to have a more powerful drug, and a population with milder natural-history drift would appear to have a less powerful one. Neither inference would be correct.

Decomposing the response separates these failure modes.

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2. Biological response (BR)

What it is

BR is the **physiological response to the drug itself.** It is what happens because of the medication's chemical action on the body, independent of belief, suggestion, or natural history. A participant who somehow received prazosin without knowing it would still develop a measurable BR; a participant who took an inert sugar pill while believing it was prazosin would have a BR of zero no matter how strong their expectation.

Methodologically, BR is the quantity that drug regulators and prescribers actually want to know. Approval decisions, dose selection, and biomarker-guided patient selection are all framed around BR. The other two components are real and inferentially relevant, but they are not the substance of pharmacological efficacy.

Real example: PTSD and prazosin

Prazosin is an alpha-1-adrenergic antagonist originally developed for hypertension. It crosses the blood-brain barrier, and at the relevant doses it antagonizes noradrenergic activity in the locus coeruleus and prefrontal cortex during REM sleep. The mechanism for nightmare reduction is reasonably specific: prazosin damps the adrenergic surge that accompanies trauma-related dream content, and the resulting suppression of REM-stage arousal reduces nightmare frequency and severity.

The pharmacokinetic profile is short. Prazosin has a half-life of two to three hours; brain concentrations equilibrate quickly with plasma. This produces a characteristic on-drug / off-drug pattern:

- **Week 1 on drug.** Brain prazosin reaches steady-state within the first few doses. Nightmare scores begin to fall. The first- week reduction is

typically small because the patient has not yet had many drug-on REM cycles to integrate.

- **Weeks 2-8 on drug.** Effect accumulates as the patient experiences several weeks of drug-on REM sleep. Under the reference calibration below, the reduction passes half its maximum at about week 5 and approaches its plateau by week 8.
- **Steady state.** Once the brain has fully adapted, additional weeks on drug produce essentially no further reduction. This is the saturating-curve property: the body has a ceiling for how much benefit any given drug can produce, and that ceiling is reached on a timescale set by the drug's own pharmacology and the disease's response time.
- **Discontinuation.** Within a few half-lives, plasma and brain concentrations return to negligible. BR drops to zero. The patient no longer has any pharmacological effect.

This pattern (saturating rise, plateau, sharp washout) is what BR captures. It is **not** linear in time; it is **not** symmetric on and off drug; and it is **not** modulated by what the patient believes. Those properties are the empirical signatures that let us identify BR separately from PB and TV.

Mathematical pattern: modified Gompertz

BR follows a saturating curve known as the modified Gompertz function, which has the form

$$BR(t) \;=\; m\,\frac{\exp(-d\,\exp(-rt)) - \exp(-d)}{1 - \exp(-d)},$$

where t here is the time on drug t_{od}. Subtracting $\exp(-d)$ gives $BR(0) = 0$, and dividing by $1 - \exp(-d)$ restores the asymptote, so $BR(t) \to m$ as $t \to \infty$ (derived; this is the form computed by `modgompertz` in `R/utilities.R`, inspected). The three parameters have direct interpretations:

Table 3: Parameters of the modified Gompertz curve for the biological response

Parameter	Symbol	Meaning
Maximum response	m	The biological ceiling. The largest reduction the drug can produce, even with infinite time on treatment.

Parameter	Symbol	Meaning
Onset rate	r	How fast the patient approaches the ceiling. High r means a fast onset; low r a slow climb.
Displacement	d	The length of the initial lag. The inflection point lies at $t = \ln(d)/r$, so a high d gives a long slow start, and a low d (near 1 or below) gives a curve that rises almost immediately. At fixed m and r a higher d does not make the rise steeper: the maximum slope, $mr/(e(1 - e^{-d}))$ for $d \geq 1$ and $mrd/(e^d - 1)$ for $d < 1$, decreases strictly in d, from mr toward mr/e. (Corrected 2026-10-08: this entry previously said a high d gives "a long slow start followed by a steep rise". See `analysis/report/compendium-mathematics.tex`, Section "Shipped parameters, landmarks, the displacement, and numerical conditioning", equation `eq:invc-maxslope`.)

For PTSD prazosin, the calibrated values shipped with the package (`data/extracted_rp.rda`, inspected) are $m \approx 11$ (10.99 points of nightmare score reduction at saturation), $r = 0.42$ per week and $d = 5$. With these values the curve inflects at about 3.8 weeks, reaches half its maximum at about 4.7 weeks (33 days) and 84 percent of it by week 8 (computed from the formula above).

The Gompertz family was chosen because it captures the 'asymptote-from-below' shape that physiological dose-response curves typically display. Two alternatives that do **not** capture this shape:

- A linear function in t ($BR = \beta_t \cdot t$) climbs without bound and is biologically implausible past a few weeks.
- An exponential approach to the ceiling ($BR = m(1-\exp(-rt))$) is acceptable for fast-onset drugs but lacks the slow-start phase that characterizes most chronic-condition responses, where receptor adaptation, feedback regulation, and downstream gene-expression changes produce a measurable lag.

The rescaled Gompertz contains the exponential approach as its low-d limit: as $d \to 0$ the ratio above tends to $1 - \exp(-rt)$ (derived). Increasing d lengthens the initial lag and moves the inflection point to $\ln(d)/r$, producing the sigmoidal slow-start shape. The three-parameter form is flexible enough to fit a wide range of pharmacodynamic profiles without overfitting (argued).

Why this matters for design

If we knew the participant's BR curve in isolation, we could read off the drug effect directly: BR at any timepoint is the answer to 'how much pharmacological reduction has accumulated by now?'. The trick of N-of-1 trial design is to construct contrasts between on-drug and off-drug timepoints that isolate BR from PB and TV. The blinded discontinuation phase, discussed in the trial-design section below, is precisely such a contrast: if the participant is taken off drug while still believing they are on drug, then BR collapses to zero while PB remains roughly intact. The difference between on-drug and off-drug response in that window is therefore a clean estimate of BR that does not require the analyst to model PB or TV explicitly.

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3. Placebo-belief response (PB)

What it is

PB is the change in symptom score that arises from the patient's **belief** that they are receiving treatment, independent of what they are actually receiving. This is the placebo response in the strict sense: the part of the improvement that would persist if the active drug were silently replaced with an inert sugar pill, provided the patient continued to believe they were on the active drug.

It is essential to be precise about what PB is and is not. PB is not 'fake' improvement, and it is not 'all in the patient's head' in any dismissive sense. The placebo response is mediated by well-characterized neurobiological pathways. Belief-driven release of endogenous opioids and dopamine produces measurable

analgesia and mood elevation. Anticipatory regulation of the hypothalamic-pituitary-adrenal axis modulates cortisol and inflammatory cytokines. Top-down attentional shifts change how symptoms are perceived and reported. Each of these mechanisms is a real biological response; what makes them 'placebo' rather than 'drug' is that they are triggered by belief rather than by pharmacology.

For trial design, the relevant point is that PB is **separately identifiable** from BR if and only if the trial design includes a phase in which the patient's belief about treatment differs from the actual treatment they are receiving. The blinded discontinuation phase of the Hendrickson hybrid design is the canonical such phase.

Real example: the blinded discontinuation contrast

Consider a sixteen-week N-of-1 trial structured in three phases:

- **Weeks 1-8 (open-label on drug).** The patient knows they are receiving prazosin. Belief in treatment is at full strength (expectancy multiplier 1.0, in the model parameterization introduced below).
- **Weeks 9-12 (blinded discontinuation).** The patient is told that, at some unannounced point during this window, they may be silently switched to placebo. They cannot reliably tell whether they are still receiving the active drug. Belief in treatment drops to a partial level (multiplier roughly 0.5) because the patient's expectation is now hedged.
- **Weeks 13-16 (open-label crossover, possibly back on drug).** The patient is told whether they are now on active drug or placebo, and belief returns to a level commensurate with that knowledge.

A patient who is silently switched to placebo at the start of week 9 will have:

- $BR = 0$ from week 9 onward (no pharmacology).
- $PB \approx 0.5 \cdot PB_{\text{open-label}}$ during weeks 9-12, reflecting hedged belief.
- TV continuing on its slow trajectory.

A patient who remains on active drug during weeks 9-12 will have:

- BR continuing at its plateau value.
- The same hedged $PB \approx 0.5 \cdot PB_{\text{open-label}}$ as the placebo-switched patient.
- The same TV trajectory.

The contrast between the two arms during weeks 9-12 is therefore a nearly-clean estimate of BR, with PB matched (because both groups have the same

belief) and TV matched (because TV is by construction independent of treatment). This is the inferential move that makes the BR-PB-TV decomposition identifiable from the data: the trial design supplies phases in which different combinations of components are present, and the analyst recovers the components by differencing those phases.

Mathematical pattern

PB also follows a Gompertz curve, but scaled by an expectancy factor $\eta(\text{phase})$ that depends on the trial phase the patient is currently in:

$$PB(t \mid \text{phase}) \;=\; \eta(\text{phase}) \cdot m_{PB}\, \exp(-d_{PB} \exp(-r_{PB} t)) + \text{constant adjustment to start at zero.}$$

Typical values of the expectancy factor:

Table 4: Typical values of the expectancy factor η by trial phase

Phase	η	Reasoning
Open-label on drug	1.0	Patient knows they are on the active drug.
Blinded discontinuation	0.5	Patient believes they may or may not still be on drug; expectation is hedged.
Open-label placebo	0.0-0.5	Patient told they are on placebo retains some residual expectation, especially if the previous active phase was beneficial.

The choice of $\eta = 0.5$ rather than $\eta = 0$ for the blinded phase is itself a modeling assumption. The empirical question is how much expectation a patient carries when uncertain about allocation, and the published placebo literature suggests it is substantial: residual expectations of 30-60% of the open-label level are reported across pain, depression, and PTSD trials (no primary source is given for this range here; it is an unverified summary and should be sourced before it is relied on). Setting $\eta = 0.5$ in the blinded phase is a reasonable default, but sensitivity analyses should vary it to confirm that conclusions are not driven by this single number.

Why expectancy variance differs by phase

A subtle property of PB is that **its variance also scales with** η, not just its mean. When the expectancy factor is high (open-label on drug), there is substantial variability across patients in how strongly the placebo response is expressed: high- suggestibility patients may show large PB, low-suggestibility patients may show essentially none, and the population variance is correspondingly large. When η is low (blinded), the placebo response is uniformly attenuated, and the patient-to-patient variance shrinks accordingly. The model parameterization reflects this with a phase-dependent standard deviation:

$$\mathrm{SD}(PB \mid \text{phase}) \;=\; \eta(\text{phase}) \cdot \sigma_{PB}^{\text{baseline}}.$$

(Inspected: `generateData` multiplies the PB standard deviation and the PB mean trajectory by the design's expectancy column `e`.)

This heteroscedastic structure is not optional: ignoring it leads to inflated standard errors in the open-label phase (where PB contributes most variance) and deflated standard errors in the blinded phase, with consequences for both confidence intervals on the drug effect and the calibration of any subsequent biomarker test.

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4. Natural history (TV)

What it is

TV is the change in symptom score that would have happened **even if the patient had received no treatment at all.** It is the participant's natural-history trajectory, integrated over the duration of the trial. For some participants TV is positive (symptoms naturally improve over the course of the trial because of life events, therapy, time-in-condition, or regression to the mean); for others TV is negative (symptoms naturally worsen because of trauma anniversaries, accumulating stress, or progressive underlying disease).

Like PB, TV is not 'noise'. It is a structured, participant-specific signal that the model can in principle estimate from data, given sufficient repeated measurements and a trial design that includes a long enough off-drug baseline or follow-up window. What distinguishes TV from PB is its origin: TV is independent of treatment status and of belief about treatment, whereas PB is modulated by both.

When does natural history actually exist?

A common simplifying assumption in trial analysis is 'subjects will improve over time, with or without treatment'. This is true for some conditions and dangerously wrong for others. The shape of TV depends on the underlying disease class, the trial timescale, and the selection process by which participants entered the trial. None of these can be assumed without thought, and the empirical shape of TV is one of the things the decomposition is designed to recover.

Acute conditions tend to TV-positive. A cold, a sprained ankle, a post-operative pain syndrome, an episode of acute back pain: these conditions resolve, on average, on a timescale of days to weeks even without intervention. A trial of ibuprofen for tension headache or of a topical analgesic for ankle sprain will see substantial TV-improvement in the placebo arm, and the apparent drug effect must be measured *against* this baseline trajectory. Failing to model TV in an acute-condition trial is a common source of inflated effect estimates.

Chronic stable conditions tend toward TV-zero on the population mean, with substantial individual variance. Type 2 diabetes under stable management, treated hypertension, controlled chronic asthma, and long-standing PTSD without active triggers are all reasonably steady on the population mean across a sixteen-week trial. But individual participants show substantial variance: one patient is doing better this month because they have started sleeping more regularly, another is doing worse because their job has become more stressful. The population mean of TV may be near zero, but the participant-level TV variance is not, and the decomposition recovers both.

Chronic progressive conditions tend toward TV-negative. Alzheimer's disease and other neurodegenerative conditions, ALS, late-stage congestive heart failure, untreated progressive multiple sclerosis, and many forms of cancer all worsen on the timescale of months. A trial of a putatively neuroprotective intervention in early Alzheimer's must explicitly model the negative TV trajectory: the apparent treatment effect is the slowing of progression relative to the worsening that would otherwise occur. Treating TV as zero in such a trial would systematically under-estimate efficacy, because the placebo arm continues to worsen and the active arm's stability looks like 'no effect' rather than 'effect is preventing the natural decline'.

Cyclical and triggered conditions show structured TV that is neither monotone nor stationary. PTSD severity often spikes near trauma anniversaries; major depressive disorder has a measurable seasonal component; relapsing-remitting multiple sclerosis oscillates on a timescale of months. In these conditions, TV is not well-described by a monotonically increasing or decreasing Gompertz curve, and the model needs either a more flexible TV specification or an explicit covariate (e.g., season, days from nearest trigger) that absorbs the cyclic component before TV estimates the residual trend.

Selection-induced TV is real and rarely zero. Even a chronic stable condition can show TV-positive average trajectory in a trial population because of how participants are recruited. Patients enroll at a moment of clinical engagement, which often coincides with a recent worsening that triggered the help-seeking behavior. Regression to the mean alone (the statistical fact that an extreme observation tends to be followed by less extreme observations on average) can produce several points of apparent TV-improvement in the early weeks of a trial, even in a population whose underlying condition is stable. This is the famous failure mode of single-arm trials in chronic-pain conditions: large 'treatment' effects evaporate when a control arm is added, because the apparent effect was almost entirely regression to the mean.

Trial participation itself induces TV. The Hawthorne effect (participants improve simply because they are being observed), the structured-visit effect (regular visits with clinicians may improve symptom management directly), and the diary-completion effect (the act of monitoring symptoms changes how they are perceived and reported) are all real, all participant-mediated, and all confounded with calendar time. They appear in the data as TV-improvement that is independent of any pharmacological or placebo response, and they apply equally to all arms of the trial. Modeling them as part of TV is the appropriate handling.

Can we assume no TV effect?

The short answer is: rarely safely, and never without empirical support. The longer answer:

- **TV may be assumed zero only if** the disease is in a stable chronic phase, the trial is short relative to the disease timescale, the recruitment process does not favor participants at peak severity, and the trial protocol does not introduce ancillary care that itself improves symptoms. For most chronic trials in psychiatric or pain conditions, at least one of these conditions fails. The conservative default is to estimate TV from the data, even at the cost of some statistical efficiency, rather than to assume it is zero.
- **The decomposition makes the assumption testable.** A model that includes a TV component will return $\hat{m}_{TV}$ and a variance estimate; if both are small relative to BR and PB, the analyst has empirical evidence for the simplifying assumption. If $\hat{m}_{TV}$ is meaningfully large, the assumption was wrong, and the trial needs the TV component to produce unbiased inference about the others.
- **Setting TV to zero and treating it as part of ε is worse than estimating it.** The participant-level TV is structured (each participant has their own slow trajectory) and not independent of the on-drug versus off-drug schedule. Lumping TV into noise produces residuals that are

autocorrelated within participants and heteroscedastic across them, and the resulting standard errors on the BR and PB estimates are wrong in both directions: too small for participants whose TV is small, too large for participants whose TV is large. Estimating TV explicitly fixes both problems simultaneously.

The general principle: TV is a *structured* component of the response, not random noise, and the decomposition treats it accordingly.

Why this matters

If TV is ignored, every TV-induced point of symptom change gets attributed to either BR or PB or to noise. This is the most common single source of bias in chronic-condition trial inference. A famous example: many early trials of chronic-fatigue therapies appeared to show large effects, until a careful natural-history arm revealed that the apparent effect was almost entirely a regression-to-the-mean phenomenon driven by the selection of patients at peak symptom severity. The patients would have improved similarly with no intervention. (This example is illustrative; no specific trials are cited for it, and it should not be quoted as documented history without a source.) TV in the model is the formal way to handle this risk.

The attribution problem is not symmetric. TV is positively correlated with the *opportunity* to detect a drug effect (chronic- condition patients are typically at peak severity at trial entry, and natural-history regression brings their scores down on average), and it is *negatively* correlated with the *credibility* of the detected effect (a regulator cannot tell, from a single-arm trial without a TV-control phase, how much of the observed improvement was treatment versus regression). Both problems are addressed by decomposing TV explicitly and by including trial phases in which BR and PB are absent.

Mathematical pattern

TV also follows a Gompertz curve but parameterized independently of BR and PB:

$$TV(t) \;=\; m_{TV}\,\exp(-d_{TV}\exp(-r_{TV}t)) \;+\; \text{offset to start at zero.}$$

The maximum m_{TV} can be either positive (natural improvement) or negative (natural worsening). Conceptually it varies across participants; in the current implementation m_{TV} is a fixed population value and participant-level variation in TV enters through the multivariate normal draw around the mean trajectory (inspected; see section 9). A trial design that aggregates over many participants can identify both the population mean and the population variance of m_{TV}, but only if there are enough off-drug timepoints to constrain the curve.

Why not just include 'week' as a covariate?

A natural objection is that a model with `week` as a covariate already controls for time. Three reasons this is not enough:

1. **Linearity assumption.** A `week` covariate enforces a linear trajectory: every additional week contributes the same fixed amount of natural-history change. Real disease trajectories are not linear; they curve, plateau, or accelerate. The Gompertz form is flexible enough to capture S-shaped or saturating patterns that linear time cannot.
2. **Participant-specific TV.** A single `week` coefficient estimates a population-mean trajectory, but participant heterogeneity in natural history is large and clinically relevant. The full BR-PB-TV decomposition uses participant- specific TV components, drawn from a population distribution, which is the correct way to express the heterogeneity.
3. **Confounding with BR onset.** Both BR and TV evolve smoothly over the same trial timescale. A `week` covariate cannot tell them apart from observational data alone; the design contrasts (on-drug vs. blinded vs. off-drug phases) are what supply identifiability. Without the explicit decomposition, the `week` coefficient absorbs both, and the analyst loses any ability to attribute the resulting time-trend.

For trials with rich within-person time structure (and N-of-1 trials are precisely such trials), the explicit Gompertz TV is the right tool, and the simpler `week` covariate is a coarse approximation that loses information.

Part II. Identification from trial design

5. How the design identifies the components

The mathematical decomposition above is meaningless unless the data contain enough information to pin down each component separately. Trial design is what supplies that information. The Hendrickson hybrid design, used as the worked example throughout the rest of this guide, has three phases that each contribute a different combination of components.

Table 5: Components present in each phase of the Hendrickson hybrid design

Phase	Drug?	Belief?	Components present
Open-label on drug	Yes	Full	$BR + PB(\eta = 1) + TV$
Blinded discontinuation	Hidden	Hedged	$BR \cdot \mathbf{1}_{\text{on drug}} + PB(\eta = 0.5) + TV$
Open-label crossover	Either	Knows	$BR \cdot \mathbf{1}_{\text{on drug}} + PB(\eta = 1) + TV$

Implementation note (inspected 2026-10-07). In the Hybrid design as coded (`analysis/scripts/figure4/04-generate-power-head.R` and the paper 13/14 driver `two-component-dgp/01-run-grid.R`), the eight occasions fall at weeks 4, 8, 9, 10, 11, 12, 16 and 20 (a twenty-week schedule, not the sixteen weeks of the narrative example), and the expectancy vector is $\eta = (1, 1, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5)$: full expectancy at the two open-label occasions and 0.5 at all six later occasions, including the final crossover, which is blinded (Hendrickson et al. 2020 describe it as a brief crossover after blinded discontinuation). The third row of the table, an open-label crossover with $\eta = 1$, and the "weeks 13-16 open-label crossover" of the example in section 3, therefore describe a variant of the design, not the coded Hybrid design. The identification argument is unaffected in kind, but in the coded design the open-label versus blinded contrast comes only from the first two occasions.

Three phase-by-allocation contrasts then identify the three components:

- **Phase 1 vs. phase 2 (within drug):** $PB(\eta = 1) - PB(\eta = 0.5) \approx 0.5 \cdot PB$. Isolates the belief component, but only approximately: the belief curve keeps rising between the compared visits (section 7.2 gives the size of the approximation).
- **Phase 2 on drug vs. phase 2 placebo:** BR alone (because PB and TV match across allocation arms within phase 2).
- **Off-drug occasions:** TV plus any residual off-drug PB, with no active BR beyond carryover (section 7.2 states this contrast in the same form).

Each contrast targets a different component, and the analyst combines them through the linear-mixed-effects model fitted to the full trajectory. The model is the formal mechanism, but the inference is driven by the design contrasts.

A trial that lacks any of these phases loses the corresponding identifiability. A pure open-label design (no blinded discontinuation) cannot separate BR from PB, because both are present at full strength throughout. A pure blinded design (no open-label phase) cannot estimate $\eta = 1$ PB, because no phase has belief at

full strength. The decomposition machinery is useful only to the extent that the design supplies identifiability, and trial designers should evaluate any proposed design by checking that each component has at least one phase or contrast in which it is uniquely present.

---

6. The Hendrickson approach

The founding pmsimstats question is the detection of a biomarker-by-treatment interaction in an aggregated N-of-1 trial. The Hendrickson reference implementation (`implementations/original/`) is the historical anchor for the whole program. Its defining feature, and the starting point for paper 06, is an asymmetry between how it generates data and how it analyzes it.

6.1 The data-generating process is fully decomposed

The DGP in `implementations/original/R/generateData.R` builds the observed symptom score as the sum of three causally distinct components, each a modified Gompertz trajectory with participant-specific random effects:

- BR (biological response, the pharmacological component), driven by the time on drug t_{od} in the current on-drug period, which restarts at zero on re-exposure (inspected in `buildtrialdesign.R`), and decaying off drug with a carryover half-life;
- PB (placebo-belief response), driven by the time-by-belief accumulation t_{pb} and scaled by the phase expectancy weight η;
- TV (natural-history component), driven by time since trial entry.

The components are summed and subtracted from baseline (lines 99 and 104 of `generateData.R`, inspected). Schematically:

```
D_t  = rowSums(t.br + t.pb + t.tv)    # total response at timepoint t
Sx_t = BL - D_t                        # symptom = baseline minus total
```

The raw simulated data retain the individual component values (`t1.br`, `t1.pb`, `t1.tv`, and so on for each timepoint), so the ground truth is three-component by construction. The prazosin-PTSD calibration used throughout the program draws its Gompertz parameters from the Hendrickson et al. (2020) dataset ($m_{BR} \approx 11$, $r_{BR} = 0.42$ per week, $d_{BR} = 5$; `data/extracted_rp.rda`, inspected).

6.2 The analysis model is a single lumped drug indicator

The analysis in `implementations/original/R/lme_analysis.R` does not mirror the DGP. It fits one continuous drug indicator, not three components:

```
Sx ~ bm + t + Dbc + bm:Dbc
random      = ~ 1 | ptID
correlation = corCAR1(form = ~ t | ptID)
```

where `Dbc` is the continuous drug state $D_{bc,it}$ (equal to 1 on drug and to $e^{-\lambda t_{sd}}$ off drug, with $\lambda = \ln 2 / t_{1/2}$), and `bm:Dbc` is the biomarker-by-drug interaction, whose coefficient is $\beta_{bm:D}$. Variant forms in the same file add an expectancy indicator (`De`), a carryover term (`tsd`) or a `bm:t` term (inspected), but they remain single-drug-indicator models. The model estimates one lumped drug main effect and one lumped biomarker slope; it does not recover BR, PB, and TV separately.

6.3 The asymmetry that paper 06 studies

The original therefore applies decomposition only as a data-generating device. It knows the response is $BR + PB + TV$ and simulates accordingly, but its inferential model collapses the three into a single treatment coefficient. This lumped estimator is exactly the baseline that paper 06 formalizes, and the central question of that paper is the consequence of the asymmetry: when does the lumped analysis of three-component data give a biased or uninformative answer, and when does adding analysis-side decomposition fix it.

7. The hybrid design

Component decomposition is meaningless unless the data contain enough information to pin down each component separately. That information is supplied by trial design, not by the analysis model. The hybrid open-label-plus-blinded-discontinuation design is the worked example throughout paper 06 and the primary identifiability vehicle.

7.1 Phase structure

The hybrid design strings together three phases, each contributing a different mixture of the three components:

Table 6: Hybrid design phase structure with drug, belief and components present

Phase	Drug	Belief	Components present
Open-label on drug	Yes	Full ($\eta = 1$)	$BR + PB(\eta = 1) + TV$
Blinded discontinuation	Hidden	Hedged ($\eta = 0.5$)	BR (on drug) $+PB(\eta = 0.5) + TV$
Open-label crossover	Either	Knows ($\eta = 1$)	BR (on drug) $+PB(\eta = 1) + TV$

The "on drug" qualifier on BR is shorthand for the continuous-decay drug state D_{bc}, not a binary switch.

7.2 The three identifying contrasts

Three phase-by-allocation contrasts identify the three components:

- **Open-label versus blinded (within drug)** yields $PB(\eta = 1) - PB(\eta = 0.5) \approx 0.5\,PB$, and so isolates the belief component, conditional on a known blinded-phase expectancy. The approximation is coarse. In the coded Hybrid design the contrast is between the on-drug visits at weeks 8 and 9, and its mean is $E(PB_8) - E(PB_9) = G(8) - 0.5\,G(9) = 2.17$ points, against $0.5\,G(9) = 2.62$, because the belief curve keeps rising between the two visits (verified at the shipped parameters, `analysis/scripts/compendium-mathematics/check-landmarks-hybrid-ceilings.R`, block 3). The contrast identifies the expectancy drop only once the curve's growth between the visits is modeled. (Added 2026-10-08. See `analysis/report/compendium-mathematics.tex`, Section "The published Hybrid construct".)
- **Blinded on-drug versus blinded placebo** yields BR once drug carryover has elapsed, because PB and TV match across allocation arms within the blinded phase.
- **Off-drug phases** yield TV plus any residual off-drug PB, with no active BR contribution beyond carryover.

The lumped LME combines these into one model, but the inference is driven by the design. The one-component analysis already exploits the drug-on/off contrast through its `Dbc` term; that is why, as section 11 shows, it is not as impoverished as its single coefficient suggests.

7.3 The expectancy identification limit

The hybrid design has one principled limitation that deserves emphasis. With only an open-label and a blinded phase, the data identify the product $\eta\, m_{PB}$, not the two factors separately. The working value $\eta \approx 0.5$ in the blinded phase is a modeling assumption, not an estimate, and absent a belief measurement it is not falsifiable from outcome data alone. The identified set for m_{PB} is the interval obtained by dividing the estimated product by η over its plausible range, collapsing to a point only when η is pinned down. Two design features close the gap:

- an open-label placebo phase, which supplies a second equation that separates η from m_{PB};
- direct measurement of treatment expectancy, as in the open-hidden and balanced-placebo designs that manipulate or record belief.

Any trial relying on the decomposition for a PB-versus-BR attribution should include such a phase or collect an expectancy measure, and should report sensitivity of the attribution to η across its plausible range.

7.4 Identifiability is necessary but not sufficient

Design-contrast identifiability does not guarantee stable estimation at finite sample sizes. The three Gompertz trajectories share an onset shape and overlap in calendar time, so their parameters can be near-collinear even when the identifying contrasts are present. Joint recovery additionally requires a phase in which each component varies while the others are held fixed, and an off-drug window long enough, relative to the carryover half-life, for TV to separate from a still decaying BR. Where these conditions are weak the components are identified only up to a near-collinear direction; the symptom is the rank deficiency that paper 06's analysis-machinery pilot encountered when the full phase-augmented formula was fitted at $N = 35$, which forced the `Dbc:phase` term to be dropped. Recoverability is to be mapped by identifiability diagnostics (convergence rates, variance-inflation factors on the component-specific parameters, and sensitivity of the BR estimate to the assumed η), not asserted.

Part III. Analysis models

8. Why the analysis model does not match the DGP

A reasonable reaction to the worked example is the following. 'If I am simulating data with three response components (BR, PB, TV) plus noise, why is the

analysis model written as `Sx ~ bm + t + Dbc + bm:Dbc` with a single random intercept and AR(1) residual correlation? Should I not include three parametric Gompertz components in the analysis, mirroring the DGP, so that the analyst sees the same structure that generated the data?' The question is reasonable, and the standard answer is that four constraints make a component-matched analysis impractical for most aggregated N-of-1 trial sizes, and that a modest extension (a phase-by-treatment indicator) captures the practically important part of the gap at minimal cost.

Four reasons not to match

1. The trial design must supply identifiability, and many designs supply only part of it. The BR-PB-TV decomposition is identified by phase contrasts: open-label versus blinded identifies the expectancy modulation of PB, on-drug versus blinded-placebo identifies BR, and off-drug timepoints identify TV. A pure open-label trial has none of these contrasts; in such a trial BR and PB both rise from zero on treatment initiation and are statistically indistinguishable from each other, regardless of how the analyst writes the model. Forcing an analysis to fit three components when the design supplies contrasts for only one or two of them does not recover the missing components; it produces unidentified or weakly identified parameter estimates with implausibly tight standard errors that misrepresent the analyst's actual inferential precision.

2. Each component term costs degrees of freedom that come out of the moderation estimand. The biomarker-moderation question is asked through the `bm:Dbc` term. A component-matched analysis adds a Gompertz BR (three parameters: maximum, onset rate, displacement), a Gompertz PB (three parameters plus a phase-specific expectancy multiplier), and a Gompertz TV (three parameters), each typically with participant-specific random effects on the maxima. That is on the order of a dozen new parameters, half of them on a non-linear scale. At trial sample sizes of $N \in [30, 150]$, the variance of $\widehat{\beta}_{bm:D}$ is dominated by participant and timepoint counts; adding parametric structure to absorb residual confounding usually inflates the moderation estimand's variance by more than it reduces bias. The linear-mixed model `Sx ~ bm + t + Dbc + bm:Dbc` with `corCAR1` residual correlation is, in this regime, the more powerful test for the question the analyst actually asks.

3. Component-matched analyses are non-linear in the parameters and convergence is fragile. Fitting Gompertz BR, PB, TV simultaneously requires either a non-linear mixed model (`nlme::nlme` with self-starting Gompertz forms), a Bayesian hierarchical fit, or a structural-equation approximation. Each is more sensitive to starting values, more prone to local optima, and more likely to fail to converge on a single trial dataset than the linear-mixed analogue. At trial-relevant N, the convergence rate of the non-linear component-matched fit is typically below 90% (an unverified estimate; no convergence rate for such

a fit is reported in this document), and the non-converging cells contaminate any aggregated analysis.

4. The moderation estimand is robust to the BR-versus-PB split for the most common scientific question. For predictive-biomarker validation, the regulator wants to know whether the biomarker predicts treatment response under the trial conditions. They typically do not require a clean pharmacological-versus-placebo decomposition of that prediction. The simple `bm:Dbc` interaction is the right marginal estimand for the validation question whether the moderation is biological or expectation-mediated. A component-matched analysis would be required only for a secondary question of the form 'is this biomarker predicting the *pharmacological* response specifically, and not the expectation response?', and that secondary question typically admits a simpler answer through phase indicators rather than through full component fitting.

A small extension that usually pays off: phase indicators

The smallest useful step toward component-aware analysis, without paying the costs above, is to add a phase indicator and its interaction with the drug indicator. Schematically,

```
Sx ~ bm + t + Dbc + phase + Dbc:phase
       + bm:Dbc + bm:Dbc:phase
       + (1 | ptID),      correlation = corCAR1(...)
```

`phase` is a categorical covariate (open-label, blinded, post- crossover) that absorbs the systematic differences in PB expectancy across phases. `Dbc:phase` tests whether the apparent drug effect shrinks under blinding, a signature that the open-label drug effect was partly PB-mediated. `bm:Dbc:phase` tests whether the moderation itself shrinks under blinding, a signature that the biomarker-by-treatment interaction was partly biomarker-by-PB rather than biomarker-by-BR.

This extension adds three to four parameters at most, well within budget at any realistic N. It does not separately estimate BR and PB as components, but it does identify the *change* in apparent drug effect that occurs when belief is hedged. For most analyses, this is the practically relevant question, and it can be answered without committing to Gompertz parametric form.

When to actually fit the full decomposition

A component-matched analysis is worth the effort when:

- Sample size is in the several hundreds (a phase 3 trial fielded as aggregated N-of-1, or a meta-analysis pooling multiple smaller trials).

- The trial design provides timepoints in every phase combination (open-label on-drug, open-label off-drug, blinded on-drug, blinded off-drug, with enough density per cell to constrain the Gompertz curves).
- The scientific question requires the BR-PB split (a pharmacology paper claiming a specific mechanism, where ruling out a placebo contribution is the central point).
- A Bayesian hierarchical fit with informative priors on the Gompertz parameters is feasible and the analyst is prepared to defend the priors.

In that regime, `lcmm` (for component-specific class-based extensions), `nlme::nlme` with self-starting Gompertz forms, or a `brms` hierarchical formulation can return participant-specific BR, PB, and TV component estimates that mirror the DGP. Outside that regime, the linear-mixed analysis with phase indicators is the better tool, and the DGP and the analysis model deliberately do not match.

The asymmetry is not a flaw of the framework; it is a deliberate consequence of the asymmetry between simulation (where the analyst controls the DGP and can specify any structure, however elaborate) and inference (where the analyst must fit a model to a finite dataset and pay the identifiability cost of every additional parameter). Simulation studies use the rich DGP to characterize how a simpler analysis behaves under realistic data generation; they do not require the analysis to match the DGP in order to produce useful inference.

---

9. The mathematics of the decomposition

The core model

For participant i at trial timepoint t:

$$Y_{it} \;=\; \mathrm{BL}_i \;-\; [\, BR_{it} + PB_{it} + TV_{it} \,] \;+\; \varepsilon_{it},$$

where the symptom score Y_{it} decreases as the components grow. Each component is a Gompertz function of its own time-on-state variable:

- BR_{it} depends on the time on drug t_{od} in the current on-drug period (it restarts at zero on re-exposure in all current DGPs), with off-drug carry-over decay.
- PB_{it} depends on the time-by-belief accumulation t_{pb}, modulated by the phase-specific expectancy factor η(phase).
- TV_{it} depends on time-since-trial-entry (calendar time).

Each component is participant-specific. In the implementation the Gompertz parameters (m, r, d) of each component are fixed, and each participant's component values at every occasion are drawn jointly from a multivariate normal centered on the Gompertz mean trajectories, with component standard deviations set separately (inspected in `generateData`). The participant-level deviations therefore enter through that joint draw rather than through random Gompertz parameters.

Joint covariance structure

The components are not independent of each other. A patient with strong pharmacological response is, on average, also somewhat more likely to develop a strong placebo expectation ('this drug is clearly working, my belief in it strengthens'), and a patient with favorable natural history may be more likely to attribute that improvement to whatever treatment they happen to be on. The Hendrickson framework parameterizes this with two cross-component correlations:

- c_{cf1t}: correlation between different factors at a single timepoint. Captures the within-time covariance of BR, PB, and TV.
- c_{cfct}: correlation between different factors at different timepoints. Captures the across-time spillover.

Together with the within-component serial correlation these define a $(2+3n) \times (2+3n)$ correlation matrix, where n is the number of occasions: an $n \times n$ block for each of the three within-component and three cross-component pairs, plus one row and column each for the biomarker and the baseline (inspected in `buildSigma`; section 17 uses the same count). The simulation draws each participant's full trajectory from a single multivariate normal of that structure.

A common confusion: variance decomposition

A frequent slip is to write

$$\mathrm{Var}(Y) \;=\; \mathrm{Var}(BR) + \mathrm{Var}(PB) + \mathrm{Var}(TV) + \mathrm{Var}(\varepsilon)$$

as if this were an alternative to working with the full covariance matrix. By the bilinearity of covariance (the variance of a sum), the correct identity for any joint distribution is

$$\mathrm{Var}(BR + PB + TV + \varepsilon) \;=\; \sum_{c} \mathrm{Var}(c) \;+\; 2 \sum_{c<c'} \mathrm{Cov}(c, c'),$$

where the second sum runs over unordered pairs of distinct terms (derived). So the simple sum-of-variances form holds if and only if the pairwise covariances sum to zero, and in particular whenever the components are uncorrelated. Under the Hendrickson parameterization with c_{cf1t} and c_{cfct} both non-zero, the components are correlated, and the cross-covariance terms are non-negligible. Working with the full covariance matrix is therefore mandatory: the sum-of-variances shorthand is a special case that does not apply to the parameter regimes used in this project.

The practical consequence is that any variance-component analysis performed on simulation output should be done after the components are extracted (using the participant-specific posterior means or variance-component estimates from the fitted mixed model), not by reading off the marginal variance of Y. The marginal variance of Y is a single number that contains no information about the internal structure.

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10. Worked example: what is lost without decomposition

Consider two participants, A and B, enrolled in a sixteen-week N-of-1 trial of prazosin for PTSD. Both are otherwise similar: same age, same baseline severity, same trial design. They differ in two latent traits that the trial does not measure directly.

- **Participant A** is a high placebo responder and a moderate pharmacological responder. Their true component values: $m^A_{BR} = 4$ (drug ceiling 4 points), $m^A_{PB} = 6$ (belief ceiling 6 points), $m^A_{TV} = 1$ (mild natural improvement).
- **Participant B** is a low placebo responder and a strong pharmacological responder. Their true component values: $m^B_{BR} = 8$, $m^B_{PB} = 1$, $m^B_{TV} = 1$.

In the open-label phase (weeks 1-8), both BR and PB run at full strength, and TV accumulates linearly. By week 8, both participants have reached close to their saturation values:

Table 7: Worked example: component values for participants A and B at week 8

	BR	$PB(\eta = 1)$	TV	Total improvement
Participant A	4.0	6.0	1.0	**11.0**
Participant B	8.0	1.0	1.0	**10.0**

A clinician examining only the totals would conclude that the two participants are responding similarly: both improved by about ten points, both look like

clinical successes. A simple t-test on the open-label change would estimate a population drug effect of about ten points and would not distinguish the two participants.

Now consider the blinded discontinuation phase (weeks 9-12). Both participants are silently switched to placebo at the start of week 9. The expectancy multiplier drops from 1.0 to 0.5. By the end of week 12, the components are:

Table 8: Worked example: component values for participants A and B at week 12

	BR	$PB(\eta = 0.5)$	TV	Total improvement
Participant A	0.0	3.0	1.0	**4.0**
Participant B	0.0	0.5	1.0	**1.5**

The two participants now look quite different. Participant A still shows a sizeable improvement (four points) under blinded placebo; participant B has nearly returned to baseline (one and a half points). This is the diagnostic contrast that the decomposition exploits.

What does each modeling strategy infer?

One-component model. A fixed-effects regression of total symptom change on phase-by-allocation indicators sees:

- Open-label means: 11.0 (A), 10.0 (B).
- Blinded-placebo means: 4.0 (A), 1.5 (B).
- Estimated drug effect: $11.0 - 4.0 = 7.0$ (A), $10.0 - 1.5 = 8.5$ (B). Population mean: 7.75.
- Estimated placebo effect: not identifiable from this contrast.
- Estimated natural history: not identifiable from this contrast.

The one-component model has produced a single number (the drug effect, or rather the mean of (BR + the drop in PB induced by discontinuation)), conflated with the placebo response that washes out under blinding. It cannot say whether the drug effect is the same for participants A and B. It estimates 'drug' at 7.75 points on average, which is an average of the true 4.0 (A) and 8.0 (B) mixed with the placebo wash-out, and it has no principled way to decompose this further.

Three-component model. A linear-mixed-effects model with the full BR-PB-TV decomposition, fitted to the same data, can recover the individual components by exploiting the differential expectancy between phases and the on-drug-versus-blinded-placebo contrast. In the idealized, noise-free limit of this illustration (these are not fitted results), the model returns:

- $\hat{m}_{BR}^{A} = 4.0$ (true 4.0). $\hat{m}_{BR}^{B} = 8.0$ (true 8.0). The drug effect is identified separately for each participant.

- $\hat{m}^A_{PB} = 6.0$ (true 6.0). $\hat{m}^B_{PB} = 1.0$ (true 1.0). The placebo response is identified, and the individual difference between A and B is preserved.
- $\hat{m}^A_{TV} = 1.0$, $\hat{m}^B_{TV} = 1.0$. Natural history is identified and matched, as it should be for participants who truly have similar trajectories.

What is at stake

Compare the inferences:

Table 9: One-component and three-component answers to clinical questions for the worked example

Question	One-component answer	Three-component answer
'Does the drug work for A?'	'Some response, magnitude unclear, ~7.75 points overall.'	'Yes: $BR_A = 4.0$ points, distinct from her larger placebo response.'
'Does the drug work for B?'	'Some response, magnitude unclear, ~7.75 points overall.'	'Yes: $BR_B = 8.0$ points, twice the magnitude of A.'
'Are A and B different responders?'	'No discernible difference; both improved by ~10 points.'	'Yes: B has 2x A's BR but 1/6 of A's PB.'
'If a biomarker predicts BR, is it useful here?'	Cannot answer (no BR estimate).	Yes if the biomarker correlates with $\hat{m}_{BR}$.
'Should A be continued on this drug long-term?'	Cannot say (the response could be largely placebo).	Probably yes (BR = 4 is real and pharmacological), but the larger placebo component will not persist if expectations decay.
'What is the population mean drug effect?'	7.75 points (biased; mixes BR and the discontinuation-induced drop in PB).	6.0 points = $(4.0 + 8.0)/2$ (correctly the mean of BR).

The one-component model has produced one biased average where the three-component model has produced six clinically actionable quantities. The information loss from lumping the components is not abstract: it makes the difference between concluding 'these patients respond similarly' and 'B has twice the pharmacological response that A does, and a biomarker predicting BR would distinguish them prospectively.' Predictive-biomarker validation, which is the whole point of running an N-of-1 trial in this design family, becomes impossible without the decomposition.

A second example: the placebo-cohort fallacy

A subtler example shows the same issue at the population level. Consider two cohorts of N-of-1 participants enrolled in methodologically identical trials of the same drug.

- **Cohort 1 (early trial, naive patients).** Mean $m_{BR} = 5$, mean $m_{PB} = 7$, mean $m_{TV} = 1$.
- **Cohort 2 (later trial, patients who have had multiple prior failed treatments and are skeptical).** Mean $m_{BR} = 5$, mean $m_{PB} = 2$, mean $m_{TV} = 1$.

By construction the drug works equally well in both cohorts ($BR = 5$ points). But the open-label totals differ substantially: cohort 1 reports thirteen-point improvements on average, cohort 2 reports eight-point improvements. A one-component analysis comparing the two cohorts would conclude that the drug 'works less well' in cohort 2 and might trigger a search for confounders, subpopulation effects, or methodological differences between the trials. None of this is real; the difference is entirely in PB, and specifically in the population baseline level of placebo responsiveness.

The three-component decomposition, applied to either cohort, returns $\hat{m}_{BR} = 5$ in both. The population difference is correctly attributed to PB, and the apparent drug-effect discrepancy disappears. This is the kind of reproducibility crisis that pharmacology has spent the last decade learning to navigate; the decomposition is one of the formal tools that makes it tractable.

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11. The paper 06 analysis

11.1 The omitted-variable-bias identity

The analytic core of paper 06 is an exact identity for what the lumped estimator estimates. By linearity of covariance, the lumped biomarker-by-treatment slope decomposes as

$$\beta_{bm}^{\text{lumped}} \;=\; \beta_{bm}^{BR} + w_{PB}\,\beta_{bm}^{PB} + w_{TV}\,\beta_{bm}^{TV},$$

so the lumped slope is displaced from the pharmacological slope β_{bm}^{BR} by a term proportional to the biomarker's covariance with PB and TV, weighted by design-dependent weights w_{PB} and w_{TV} (derived in paper 06). The displacement is in the probability limit, not the variance: a larger one-component trial converges on the same biased value. The bias is zero if and only if the

weighted sum $w_{PB}\,\beta_{bm}^{PB} + w_{TV}\,\beta_{bm}^{TV}$ vanishes, which holds in particular when the biomarker is orthogonal to PB and TV, regardless of the magnitude of those components. The companion identity for the lumped treatment effect is displaced by the PB and TV contributions themselves, so a lumped pre-post treatment effect is biased whenever PB or TV is non-zero.

11.2 Three analysis strategies compared

Paper 06 compares three analysis strategies against three-component data:

1. **One-component (lumped).** The Hendrickson model, `Sx ~ bm + t + Dbc + bm:Dbc`.
2. **Phase-augmented LME.** A modest extension, `Sx ~ bm + t + Dbc + phase + Dbc:phase + bm:Dbc + bm:Dbc:phase`, which tests whether the drug and biomarker-by-drug effects attenuate under blinding, without the full Gompertz parametric form.
3. **Full component-matched decomposition.** A parametric model that mirrors the DGP and recovers component-specific slopes, feasible only inside the recoverable region of section 7.4.

11.3 What the simulations show

Naming note (2026-10-07). Paper 06 now labels its studies Study A, a contamination pilot, and Study B (recovery under a belief-decoupling design). Earlier versions of this document called the last two Study B and Study C; the labels below follow paper 06. All N are totals across randomization paths (NOTATION.md, rule 4).

Study A (orthogonal-biomarker arm; 1,000 replicates per alternative cell, 5,000 per null). With the biomarker coupled to BR only ($c_{bm,PB} = 0$), the one-component estimate of the biomarker-by-treatment slope is essentially unbiased across the entire (m_{PB}, m_{TV}) grid: mean absolute bias 0.016 at $N = 35$ and 0.006 at $N = 300$ against a true value $\beta_{bm:D} = -2.25$, both within one Monte Carlo standard error of zero, with near-nominal type I error (0.031 to 0.056 for the one-component analysis). The phase-augmented analysis reduces no bias (0.028 versus 0.016 at $N = 35$) and is markedly less powerful (mean power 0.35 for the phase-augmented versus 0.59 for the one-component analysis at $N = 35$, and 0.62 versus 0.90 at $N = 70$; both reach approximately 1.0 by $N = 200$). Under an uncontaminated biomarker, decomposition is therefore strictly dominated. (Simulated; verified on 2026-10-07 against `study-a-summary.rds` in the `study-a-alt1000-null5000` directory of `analysis/data/derived_data/component-decomposition/`. Earlier versions quoted the $N = 300$ and $N = 200$ cells as $N = 150$ and $N = 100$, which were per-path counts before the 2026-08-08 migration to totals.)

Contamination pilot (coupling arm; 100 replicates per cell). With the biomarker coupled to PB, the one-component bias scales with the coupling strength $c_{bm,PB}$, not with the magnitude of PB, as the identity predicts (the one-component bias reaches approximately +0.51 at $c_{bm,PB} = 0.45$, $N = 300$; simulated, as reported in paper 06). The phase-augmented analysis offers no protection: its bias is statistically indistinguishable from the one-component bias at every contamination level.

Study B (recovery under a belief-decoupling design; 1,000 replicates per cell, $N \in \{70, 150\}$, $c_{bm,PB} \in \{0, 0.1, 0.2, 0.3\}$). This is the decisive design comparison. On the standard hybrid design the drug-state and belief contrasts are collinear (analysis-stage correlation approximately +0.6), and the belief-covariate, blinded-stratum, and Gompertz-basis decompositions all fail to recover β_{bm}^{BR}, no better than the one-component analysis. A balanced-placebo (open-hidden) design adds a covert on-drug phase ($\eta = 0$) and an open-placebo phase ($\eta = 1$), reducing the drug-state-by-belief correlation to approximately −0.66 and breaking the collinearity. On that single-path design the belief-covariate decomposition returns the unbiased pharmacological slope across the feasible contamination range (bias −0.022, +0.008, +0.021, −0.026 at $N = 150$, MCSE approximately 0.02; coverage 0.94 to 0.96), whereas the one-component estimator degrades monotonically (bias +0.16, +0.34, +0.48; coverage down to 0.86). (Simulated; verified on 2026-10-07 against `study-b-balanced-summary.rds` in the same directory tree.) The grid is capped at $c_{bm,PB} = 0.30$ because beyond approximately 0.45 the implied covariance is non-positive-definite.

The power comparison between the two estimators on the balanced-placebo design is more nuanced than a uniform decomposition penalty (rejection rate for the interaction at $\alpha = 0.05$, computed from the 1,000-replicate per-cell p-values; paired standard error of the gap at most 0.014; power values verified against the same summary file):

Table 10: Study B (recovery) power of one-component and decomposition analyses on the balanced-placebo design

$c_{bm,PB}$	N	One-component	Decomposition	Gap
0.0	70	0.85	0.66	+0.19
0.1	70	0.78	0.67	+0.11
0.2	70	0.74	0.69	+0.05
0.3	70	0.64	0.67	-0.03
0.0	150	0.99	0.93	+0.07
0.1	150	0.98	0.93	+0.05
0.2	150	0.97	0.93	+0.04
0.3	150	0.93	0.93	-0.00

At zero contamination the one-component analysis is the more powerful of the

two (by 0.19 at $N = 70$, 0.07 at $N = 150$), reflecting the degrees of freedom the decomposition spends. That advantage erodes monotonically as contamination grows and reverses by $c_{bm,PB} = 0.3$ (0.64 versus 0.67 at $N = 70$), because contamination in this direction drags the biased one-component estimate toward the null and so costs it power, while the decomposition holds the estimate at the pharmacological target. The one-component power advantage is therefore confined to the low-contamination regime in which decomposition is unnecessary. (The sign of the gap depends on the direction of the coupling: a coupling that reinforced the BR sign would inflate the one-component estimate and give it spuriously higher power, which is power to detect a contaminated quantity rather than a genuine advantage.)

On the hybrid design the same comparison is uninformative because power saturates near 1.0 for both estimators (the BR effect of -2.25 is large relative to its standard error at the two-path hybrid totals $N = 140$ and $N = 300$ of the recovery runs; inspected in `study-b-recovery-summary.rds`), and both are biased under contamination in any case; there the operative axis is bias and coverage, not power.

12. When decomposition pays off

The guidance of this section and of section 13 was written before the reductions of Part V were evaluated. Section 21 revises it. Where the two differ, section 21 governs.

The inferential value of decomposition is conditional, not universal, and the condition is a joint function of the biomarker's correlation structure and the trial design, not of either alone. Three points summarize the result.

1. **The design contrast matters, not the parametric model.** The trial must supply the drug-on/off and belief contrasts that identify the components. The lumped one-component analysis already exploits the drug-on/off contrast through `Dbc`. Adding parametric component structure on top is not free: it spends degrees of freedom and, under an uncontaminated biomarker, reduces no bias while losing substantial power.

2. **For the biomarker-by-treatment slope, the bias is governed by covariance, not magnitude.** A biomarker orthogonal to PB and TV gives an unbiased lumped slope no matter how large PB and TV are. A biomarker coupled to PB or TV gives a contaminated slope that no amount of data fixes, because the failure is identification, not precision.

3. **Recovery under contamination is a design property.** When the biomarker is coupled to PB, the bias is recoverable only under a design that varies belief independently of drug exposure. On the standard hybrid design no analysis recovers it; on a balanced-placebo design the belief-covariate decomposition does.

The bottom line specific to the hybrid design

For the biomarker-by-treatment interaction estimand on the hybrid design, component decomposition never wins:

- if the biomarker is orthogonal to PB and TV, the lumped analysis is already unbiased and is the more powerful of the two, so decomposition only costs power;
- if the biomarker is coupled to PB, the lumped analysis is biased but decomposition cannot rescue it on the hybrid design, because the identifying contrasts are collinear.

Decomposition earns its place on the hybrid design only for a different estimand: the **attribution** of response to pharmacology versus belief versus natural history. For attribution the hybrid design is the primary identification vehicle, subject to the η limit of section 7.3 and the collinearity caveat of section 7.4.

Was the decomposition in Hendrickson's DGP unnecessary?

A natural reading of the paper-06 result is that, because the lumped analysis suffices for the biomarker-by-treatment estimand, the three-component structure Hendrickson built into the simulator was itself superfluous. That reading conflates two distinct choices. The decomposition was unnecessary in the *analysis* model; it was load-bearing in the *data-generating process*. The two claims are not interchangeable.

The DGP decomposition was necessary, or at least not removable, for two reasons.

- It defines the ground truth the whole program tests against. The omitted-variable-bias identity of section 11.1, and every simulation that confirms it, can be exhibited only if the simulator generates BR, PB, and TV as separately controllable channels. A DGP that produced a single lumped treatment effect would offer no $c_{bm,PB}$ knob to turn, no way to construct the contamination, and therefore no way to show that the lumped analysis is unbiased under orthogonality but biased under coupling. The DGP decomposition is what makes the bias question askable.
- It encodes genuinely distinct mechanisms with different time-drivers: BR by cumulative time on drug with off-drug carryover decay, PB by time on belief scaled by the phase expectancy η, and TV by time since entry. These produce different trajectories across trial phases, and the design-contrast structure of section 7.2 (open-label versus blinded yielding approximately $0.5\,PB$, blinded on-drug versus blinded placebo yielding BR) cannot be reproduced from a single lumped curve. The components are not cosmetic; they drive the phase behavior the whole identification argument relies on.

What paper 06 shows to be unnecessary is only the symmetric move of mirroring that structure in the analysis. Fitting the components back out is unnecessary, and under an orthogonal biomarker counterproductive, for the biomarker-by-treatment estimand, and unrecoverable on the hybrid design when the biomarker is coupled. Hendrickson's choice to analyze with a single lumped `Dbc` was, in hindsight, the appropriate one for that estimand.

The qualifier is the estimand. The analysis-side lumping is validated as sufficient only for the interaction-slope target under an orthogonal biomarker. Had the scientific target been attribution, or a PB-coupled biomarker, the lumped analysis would be insufficient, and the DGP's decomposition would then require both a matching analysis and a belief-decoupling design. The accurate summary is therefore that the DGP decomposition was necessary to pose and answer the question, whereas the analysis-side decomposition was unnecessary for the specific estimand Hendrickson actually pursued.

13. Practical advice for the hybrid design

The hybrid design is the right default for aggregated N-of-1 trials whose primary estimand is the pharmacological treatment effect or a biomarker-by-treatment interaction with a mechanistically pharmacological biomarker. It is not the right design when the biomarker may couple to placebo responsiveness and the pharmacological slope must be recovered. The following guidance operationalizes that.

Use the hybrid design, with the lumped one-component analysis, when:

- the estimand is the lumped treatment effect or a biomarker-by-drug interaction, and
- the candidate biomarker is mechanistically orthogonal to belief and to natural history. Pharmacokinetic or receptor-binding biomarkers (for example a CYP metabolizer phenotype, or an adrenergic-receptor variant affecting prazosin binding) qualify: enzymatic metabolism and receptor affinity cannot plausibly couple to belief state or untreated disease trajectory. Here the lumped slope is already the target, and a parametric component model is counterproductive.

Use the hybrid design, but add an open-label placebo phase or an expectancy measure, when:

- the estimand is attribution (how much of the response is pharmacological versus belief versus natural history). The base two-phase hybrid identifies only the product $\eta\, m_{PB}$; the open-label placebo phase or a recorded expectancy separates them. Report attribution sensitivity to η across its

plausible range, and verify the off-drug window is long enough, relative to the carryover half-life, for TV to separate from a decaying BR.

Do not rely on the hybrid design; field a balanced-placebo (open-hidden) design instead, when:

- the estimand is the pharmacological biomarker slope and the candidate biomarker may correlate with placebo responsiveness or with natural-history drift. Psychological or symptom-severity biomarkers fall here: baseline PCL-5 severity, for example, couples to both expectancy and regression-to-the-mean TV, and a COMT genotype is a documented predictor of placebo-response magnitude. On the hybrid design the contamination is unrecoverable; only the covert-drug and open-placebo phases of the balanced-placebo design decouple belief from drug exposure enough to recover the unbiased slope.

Design-stage checklist before committing to the hybrid design.

- Identify the estimand first: lumped effect, attribution, or pharmacological biomarker slope. The estimand, not convenience, selects the design.
- Classify the biomarker as orthogonal to or potentially coupled with PB and TV, on mechanistic grounds, before enrollment. If coupling cannot be ruled out, the hybrid design is the wrong instrument.
- Verify that every component the estimand requires has at least one phase or contrast in which it is uniquely present. A pure open-label design cannot separate BR from PB; a pure blinded design cannot reach full-strength PB.
- If attribution is in scope, plan the open-label placebo phase or expectancy measurement at design time, and size the off-drug window against the carryover half-life.
- Do not plan to buy bias protection with the analysis model. On the hybrid design the phase-augmented and full-decomposition analyses do not reduce biomarker-slope bias; design is the only lever.

Part IV. The decomposition in the wider literature

Parts I through III have developed the decomposition on its own terms. We turn now to the question of how it sits in relation to the clinical trials and N-of-1 literatures, since a reader deciding whether to adopt it, or which of its components to carry, will want to know what is conventional and what is not.

We should say at the outset that the three components are not equally well established. One of them is standard, one is contested, and one is an innovation of this program. That asymmetry turns out to matter for the reductions considered in Part V, and it is the reason this part precedes them.

14. Placebo response and the three-arm tradition

The separation of placebo response from natural history has a long and contentious history in clinical trials methodology, and it is worth recalling why.

A two-arm trial compares active treatment against placebo. The difference between the arms estimates the pharmacological effect, which is what regulators want. But what does the placebo arm itself measure? It is tempting to read improvement in the placebo arm as the placebo response, that is, as the benefit conferred by the belief that one is being treated. This reading is mistaken, or at least unsupported, because the placebo arm also contains everything that would have happened anyway. Regression to the mean, spontaneous remission, seasonal variation, and the ancillary care that trial participation itself supplies are all present in the placebo arm and none of them is a placebo effect.

Separating the two requires a third arm in which participants receive no treatment at all. Hróbjartsson and Gøtzsche assembled the trials that had done this, and their conclusion was deflationary. Across their reviews, the last covering 202 trials in 60 conditions, they found little evidence of a placebo effect on objective outcomes, and what remained on subjective outcomes was modest. Much of what the field had called placebo response was, on their reading, natural history and regression to the mean wearing a different name.

We do not need to adjudicate that debate here. The relevant point for present purposes is narrower. The empirical literature regards the belief channel and the natural-history channel as separable only under a design that most trials do not field. A simulation that instantiates both as distinct latent trajectories is therefore asserting an identification which is, at best, difficult to obtain in practice.

This bears directly on the hybrid design of section 7. That design supplies an open-label and a blinded phase, and the contrast between them identifies the product of the expectancy weight and the placebo-belief maximum. It does not supply a no-treatment arm. As section 7 sets out, the working value of the expectancy weight in the blinded phase is a modeling assumption rather than an estimate.

15. Trend terms in the N-of-1 and crossover literature

The natural-history component stands on quite different ground.

A time trend is a standard element of the crossover and N-of-1 model. Period effects, drift, and serial correlation are routine in that literature, and the reason is straightforward. A within-person design measures the same participant repeatedly over weeks or months, and any process that moves the outcome over that span will be confounded with the treatment schedule unless it is modeled. Senn treats period effects as a first-order concern throughout his work on crossover trials. Schmid and Yang's Bayesian hierarchical formulation for N-of-1 data carries an individual-level trend parameter alongside the treatment effect and the autocorrelation.

The natural-history component of this decomposition is the formalization of that trend. It is more elaborate than a linear drift term, since it takes a Gompertz form and carries participant-specific random effects, but it occupies the same position in the model and answers the same concern.

We note one consequence which is easy to overlook. The analysis model fitted throughout this program includes a linear term in time. That term is the analysis-side counterpart of the natural-history component. There is no corresponding term for placebo-belief, since the expectancy regressor `De` exists in the software but is disabled (`useDE = FALSE`) in the drivers of papers 06, 13 and 14 and of the Figure 4 reproduction (inspected; several other drivers, for example those of papers 03, 05 and 09, set `useDE = TRUE`). For the decomposition papers the decomposition is therefore matched to the analysis model in one component and not in the other.

16. What this program adds, and what it assumes

A modeled placebo-belief trajectory, with its own Gompertz parameters and its own participant-level random effects, is not inherited from either literature. It is an innovation of the Hendrickson framework and of this program, and it is worth being explicit about what it buys and what it costs.

On the plus side, it makes the attribution question askable. A trial that can separate pharmacological response from expectancy response can ask whether a candidate biomarker predicts the former or merely the latter, and that is a question with real regulatory and clinical consequence. Part III's worked example shows what is lost when the question cannot be posed.

On the minus side, the identification rests on the expectancy weight, which the design does not identify. It also asserts a separation which, as section 14 records, the empirical placebo literature regards as attainable only under designs with a no-treatment arm, which this program does not simulate. The assumption is not unreasonable, but it is an assumption and it should be labeled as one.

The reader may reasonably ask how much of the framework depends on it. That is precisely the question Part V takes up, by removing one component at a time and observing what changes.

Part V. The two-component reductions

This part is new. It reports and interprets papers 13 and 14, and where it disagrees with Parts I to III the disagreement is flagged explicitly rather than smoothed over.

17. The question

The three-component decomposition is expensive. At eight measurement occasions it makes the correlation matrix $(2 + 3n) = 26$ square, with three within-component blocks and three cross-component blocks, and it imposes positive-definiteness constraints that bound the biomarker-moderation parameter.

For a study whose object is the *comparison* between data-generating architectures, or between analysis specifications, it is fair to ask whether all three components earn their place. Two reductions to two components are available:

Table 11: Components retained and dropped by the two-component reductions of papers 13 and 14

paper	components retained	dropped
13	BR + TV	placebo-belief (PB)
14	BR + PB	natural history (TV)

Both take the matrix to $(2 + 2n) = 18$ square with one cross-component block. Both reproduce paper 01's architecture comparison: covariance moderation loses power as carryover grows, mean moderation does not, and the crossover design is insensitive to carryover under either. **Neither reduction changes the substantive conclusion**, so the choice between them is not a choice about results. It is a choice about which assumption to carry.

Note (2026-10-07). The architecture comparison referred to here is the version of paper 01 that papers 13 and 14 reproduced. Paper 01 has since been revised. The published step coupling makes the coupling gap zero, so the interaction vanishes at any positive half-life; graded coupling, $g_t = 2^{-t_{sd}/t_{1/2}}$ off drug and 0 before exposure, removes that collapse (docs/37). Under graded coupling, covariance-based binding and mean moderation that fades with the drug effect lose power almost identically (paper 01, abstract). The two-and-three-component comparison in this part is unaffected, but the contrast "covariance loses power, mean does not" should not be quoted as the current paper 01 result.

18. The four axes

The reductions differ on four axes, and they do not all point the same way.

Table 12: The TV-drop and PB-drop reductions compared on four axes

axis	TV-drop (paper 14)	PB-drop (paper 13)	favors
Safety	unconditional	conditional on `c.bm.pb = 0`	14
Design structure	nothing becomes inert	expectancy weight inert	14
Analysis-model match	`t` term fits nothing; PB unmodeled	DGP matches the fitted model	13
Convention	deletes the conventional trend term	matches N-of-1 practice	13

18.1 Safety

The biomarker cannot couple to TV. The implementation provides an optional biomarker-PB correlation (`c.bm.pb`), used in paper 06's contaminated-biomarker study, but **there is no biomarker-TV counterpart anywhere** in the package, the drivers, or any implementation collection. The biomarker-TV entry of the correlation matrix is identically zero under every parameter setting the software can express (inspected: no `c.bm.tv` or equivalent argument exists in `R/`, `implementations/` or `analysis/scripts/` as of 2026-10-07).

By paper 06's omitted-variable identity, the one-component biomarker slope is

$$\beta_{bm}^{\text{lumped}} \;=\; \beta_{bm}^{BR} + w_{PB}\,\beta_{bm}^{PB} + w_{TV}\,\beta_{bm}^{TV},$$

so a component contributes bias only through the biomarker's covariance with it. Deleting a block whose coupling is *structurally* zero cannot change the estimand. Deleting a block whose coupling is merely *set* to zero changes the estimand as soon as someone sets it otherwise.

Paper 14's reduction is of the first kind, paper 13's of the second. A study setting `c.bm.pb` nonzero cannot use paper 13's DGP at all.

Caveat that matters. Paper 14's safety rests on a *gap in the software*, not on a property of natural history. Under the same identity, a biomarker correlated with natural-history drift would displace the slope exactly as a PB-correlated biomarker does. Closing that gap, by adding the missing parameter, would both test the untested half of paper 06's central claim and tell us how much paper 14's advantage is really worth.

18.2 Design structure

The design's expectancy weight η, which distinguishes open-label from blinded occasions, enters the data-generating process at exactly two points, both inside the PB component: it scales the PB mean trajectory and the PB standard deviation. It appears nowhere else.

With PB removed, η has no effect on the generated data. The open-label versus blinded contrast becomes invisible to the simulation, and the three trial designs are distinguished only by their on-drug and off-drug patterns.

The mirror is not true. Dropping TV renders **nothing** inert: the calendar-time argument that drives the TV mean also drives the AR(1) week gaps of every within-component block, so unequal visit spacing survives, and η continues to act through PB.

This asymmetry is one-sided and it favors paper 14.

Mitigation, from Part II. Section 7.3 above establishes that the hybrid design identifies only the product $\eta\, m_{PB}$, and that the working value $\eta \approx 0.5$ is a modeling assumption rather than an estimate, not falsifiable from outcome data without an expectancy measurement. If η is not identified in the first place, paper 13's cost in rendering it inert is smaller than it first appears. The cost is real for any study that *manipulates* expectancy, as the balanced-placebo designs do, and largely notional for a study that merely carries a blinded phase.

18.3 Analysis-model match

This axis was not visible until the analysis model was read alongside the DGP, and it reverses the balance of the first two.

The fitted model is assembled in `R/lme_analysis.R` (inspected; schematic) as

```
modelbase <- "Sx ~ bm + t"          # always
... + Dbc + bm:Dbc                  # when drug status varies
... + De                            # only when useDE = TRUE
```

Both papers' driver (`analysis/scripts/two-component-dgp/01-run-grid.R`) sets `useDE = FALSE` (inspected). So:

- **TV has a fitted analysis-side counterpart**, the `t` term, present in every fit.
- **PB has none** in these runs. The expectancy regressor `De` exists but is disabled.

The consequence is that the two reductions differ in how well the DGP matches what the analysis actually estimates:

- **Paper 13 (BR + TV).** The DGP generates TV, and the analysis fits a trend term that partially absorbs it. The DGP generates no PB, and the analysis models no PB. The two are coherent.
- **Paper 14 (BR + PB).** The DGP generates PB, and the analysis models it nowhere, so PB is lumped into the residual. The DGP generates no TV, and the analysis fits a `t` term with nothing to fit.

Part I warns specifically against the first of those: lumping a structured participant-level component into noise produces residuals that are autocorrelated within participants and heteroscedastic across them, with standard errors wrong in both directions. Paper 14 does this to PB.

The effect did not show up as a Type I error problem in either paper's results, which is reassuring but not decisive: both papers' null cells show the same design-dependent calibration pattern, so any additional distortion from unmodeled PB is not separable from it in the runs performed.

18.4 Convention

A time trend is a standard element of crossover and N-of-1 models. Period effects, drift, and serial correlation are routine, and the TV component is the formalization of that trend. A modeled placebo-belief trajectory with its own Gompertz parameters is an innovation of this program rather than an inherited convention; the wider literature handles placebo response by blinding.

The placebo literature reinforces the point. Hróbjartsson and Gøtzsche's systematic reviews, the last covering 202 trials across 60 conditions, find that much of what is attributed to placebo response is not separable from regression to the mean and natural-history drift without a no-treatment arm. Their three-arm methodology exists because the two channels are hard to distinguish empirically. A simulation instantiating both as separate latent trajectories asserts an identification the empirical literature regards as attainable only under designs most trials do not field.

On this axis paper 13's model is the more familiar and the less committed, and paper 14's is neither.

19. Conflict with section 4

Section 4 contains a passage headed *"Can we assume no TV effect?"* whose answer is "rarely safely, and never without empirical support." It lists four conditions under which TV may be set to zero and observes that "for most chronic trials in psychiatric or pain conditions, at least one of these conditions fails." It then states that setting TV to zero and treating it as part of the residual "is worse than estimating it."

Paper 14 does what that section warns against, in a psychiatric indication. The conflict is real and should not be smoothed over.

Two things reduce but do not eliminate it.

First, Part I argues about an *analysis* that omits TV from data which contains it. Paper 14 removes TV from the *data-generating process*, so there is nothing to omit and no residual structure to mis-model. The specific harm Part I names, autocorrelated and heteroscedastic residuals, does not arise.

Second, paper 14 is explicitly a methodological device rather than a clinical model, and its stated scope is the comparison between architectures. Part I's warning is addressed to analysts of real trials.

What survives the two qualifications is this: **paper 14's DGP is not a defensible model of a chronic psychiatric indication**, and should not be described as one. On this program's own taxonomy PTSD is a cyclical and triggered condition whose natural-history trajectory is "structured and non-stationary," with selection at moments of clinical engagement contributing regression to the mean on top. A simulation that deletes TV cannot represent any of that.

Paper 14's limitations section states this. Readers should not carry its power figures into any claim about what a real prazosin trial would achieve.

20. What the reductions do not buy

Both papers expected the reduction to relax the positive-definiteness constraint on the moderation parameter. It does not, and the reason is analytic rather than empirical.

The constraint is a Schur complement on the biomarker row. Writing M for the component block and v for the unit-scaled coupling pattern, the correlation matrix is positive definite if and only if $1 - c_{bm}^2 \, v^\top M^{-1} v > 0$, so the exact ceiling is

$$c_{bm}^{*} \;=\; \left(v^\top M^{-1} v\right)^{-1/2}$$

(derived; papers 13 and 14, Appendix C). Under compound symmetry, with common within-component correlation ρ, same-occasion cross-component correlation c_1 and different-occasion cross-component correlation $c_\times$, M is a sum of Kronecker products (papers 13 and 14 call it a Kronecker sum) and its eigenvalues collapse to those of two small reductions, giving a smallest eigenvalue of

$$\lambda_{\min}(M) \;=\; (1 - \rho) - (c_1 - c_\times)$$

(derived; checked here by direct block-eigenvalue argument), which contains **neither the number of occasions nor the number of components**. This holds when there are at least two components ($K \geq 2$), $c_\times \leq \rho$ and $c_1 \geq c_\times$; with $K = 1$ the smallest eigenvalue is $1 - \rho$, with $c_\times > \rho$ the eigenvalue $[1 + (n-1)\rho] - [c_1 + (n-1)c_\times]$ is smaller, and with $c_1 < c_\times$ the eigenvalue $(1 - \rho) + (K-1)(c_1 - c_\times)$ is smaller (for example 0.45 rather than 0.85 at $\rho = 0.1$, $c_1 = 0.2$, $c_\times = 0.15$, $K = 3$, $n = 8$; and 0 rather than 0.45 at $\rho = 0.7$, $c_1 = 0.05$, $c_\times = 0.2$). At the program's reference values ($\rho = 0.7$, $c_1 = 0.2$, $c_\times = 0.1$) the conditions hold. (Corrected 2026-10-08: the formula was previously stated without these conditions. Verified, `analysis/scripts/compendium-mathematics/check-landmarks-hybrid-ceilings.R`, block 7. See `analysis/report/compendium-mathematics.tex`, Section "The number of response components and the ceiling".) Deleting a third of the rows and columns does not touch the binding structural constraint. (The symbol $\lambda_{\min}$ denotes an eigenvalue here, not the carryover decay rate λ.)

Numerically the reduction buys between 0.007 and 0.013 of feasible c_{bm} range across designs and half-lives (computed in paper 14; corrected 2026-10-08 from 0.008: the smallest gain is OL+BDC at $t_{1/2} = 1$, $0.540 - 0.533 = 0.007$, unrounded 0.0069, verified in `analysis/scripts/compendium-mathematics/check-landmarks-hybrid-ceiling` block 8). By contrast, the number of occasions matters far more: holding the design shape fixed, the three-component ceiling falls from approximately 0.81 at $n = 4$ to 0.58 at $n = 12$ (paper 14, Appendix C). An earlier version of this sentence said that halving the number of occasions moves the ceiling by roughly 0.2; paper 14 does not report a halving comparison, so the figures it does report are given instead. These ceilings belong to the paper 13 and 14 parameter set and are not comparable with the compound-symmetry ceiling of 0.256 or the separable AR(1) ceilings (0.440 at $\rho = 0.8$; 0.474, Hybrid 0.480, at $\rho = 0.7$) of paper 01.

The case for either reduction is transparency and cost, not numerical headroom. Anyone adopting one for the latter reason has misdiagnosed the constraint.

21. Revised guidance

Superseding the guidance of Parts I and III where they differ.

Use the full three-component decomposition when

- the estimand is the attribution of response to pharmacology versus belief versus natural history;
- a candidate biomarker may correlate with a non-pharmacological component;

- the study manipulates or measures expectancy, including any balanced-placebo or open-hidden design;
- absolute power is reported as a design input rather than as a comparison;
- the simulation is offered as a model of a specific clinical indication.

A two-component reduction is defensible when

- the object is a comparison among architectures, analysis specifications, designs, or inference procedures;
- absolute power figures are used only relative to one another;
- the reduction and its costs are stated rather than assumed.

Choosing between the two

Neither dominates; the axes are split two and two.

- **Prefer dropping TV (paper 14)** for work internal to this program where the contaminated-biomarker case is live, or where the blinding contrast is used, since it is safe under every parameter setting and costs no design structure.
- **Prefer dropping PB (paper 13)** for work intended for the wider N-of-1 methodological community, since it presents a familiar model, avoids asserting a contested decomposition, and is the better matched to the analysis model actually fitted.
- **Prefer neither for clinical modeling** of a chronic psychiatric indication.

What should not happen is that either reduction be adopted without noticing that a choice was made.

Flattening versus removing

A third option exists and is distinct from both. Setting a component's Gompertz maximum to zero flattens its mean trajectory while leaving its variance contribution and its covariance blocks in place, because the implementation sets component standard deviations independently of the maximum. For PB, flattening also preserves η's effect on the PB standard deviation, so a flattened model **retains the blinding contrast that a removed model loses**.

A study wanting the simpler mental picture without the change in residual variance, or without losing the design structure, should flatten rather than remove.

22. Open problems

Two sets of open items are collected here. The first is inherited from the hybrid-design analysis of Part III, and the second arises from the reductions of the present part.

22.1 Inherited from the hybrid-design analysis

The Study B recovery result (paper 06's current label; called Study C in earlier versions of this document) is established at 1,000 replicates over $c_{bm,PB} \in \{0, 0.1, 0.2, 0.3\}$ and $N \in \{70, 150\}$; recovery beyond the positive-definite range ($c_{bm,PB}$ above approximately 0.45), participant-level component recovery, and replication under a mean-moderation coupling remain open. The contamination-pilot evidence is a 100-replicate pilot; its production run is set out in the study's working protocol (`analysis/scripts/component-decomposition/00-ademp-protocol.md`).
The η working value of 0.5 in the blinded phase remains a modeling assumption absent a belief measurement. The framework's participant-level claims are conditional on falling inside the recoverable region of section 7.4.

22.2 Arising from the reductions

1. **Add `c.bm.tv`.** The absence of a biomarker-TV coupling parameter is why paper 06's central claim has been demonstrated for PB and only asserted for TV, and it is why paper 14's safety argument rests on a software gap. Adding it would close both.
2. **Run a matched three-component grid** under the corrected carryover recursion, so that the power difference between two and three components can be decomposed into its variance and recursion parts. Neither paper can currently do this.
3. **Diagnose the design-dependent Type I error.** Both papers find pooled rejection rates running from roughly 0.06 under CO to 0.02 under OL+BDC against a nominal 0.05, architecture-independent and monotone in how much off-drug information the design supplies. Neither diagnosed it. Whether the cluster-robust correction that removes a milder conservatism in paper 02 also flattens this pattern is untested.
4. **Test whether unmodeled PB degrades inference** in paper 14's configuration, which Section 18.3 identifies as a theoretical concern that the existing runs cannot separate from the Type I pattern of item 3.

Part VI. Reference

23. Common questions

Q: Could I just use a random intercept and random slope per participant?

A random intercept-and-slope mixed model is

$$Y_{it} \;=\; (\beta_0 + u_{0i}) + (\beta_t + u_{1i})\, t + \varepsilon_{it},$$

with participant-specific intercept deviation u_{0i} and slope deviation u_{1i} (the symbols α and β_i used in earlier versions were renamed because α is reserved for the significance level). This captures heterogeneity in baseline severity and in linear time trends, but it cannot decompose the time trend into BR-style saturating-drug, PB-style expectancy-modulated, and TV-style natural-history pieces. It also forces each participant's trajectory to be linear in t, which is biologically wrong for chronic-condition responses on the timescale of weeks. The BR-PB-TV decomposition is more specific about *why* the trajectory has the shape it does, and that specificity translates directly into identifiability of the clinically distinct components.

Q: Isn't TV just disease natural history? Why not measure it independently?

In principle yes. In practice, three obstacles:

1. **Ethical.** Withholding active treatment from patients with active symptoms to measure natural history is rarely acceptable.
2. **Practical.** Natural-history studies require long observation windows in the absence of intervention, and they typically lack the within-person matching that an N-of-1 design naturally provides.
3. **Statistical efficiency.** When the same trial collects on- drug and off-drug data on the same participants, the model extracts TV from the off-drug timepoints with high efficiency, and a separate natural-history study is redundant.

The decomposition is the way to recover TV from the trial itself, without paying the cost of a parallel natural-history study.

Q: Why is the biomarker only correlated with BR, not with PB?

This is a modeling assumption, not a fact about reality. The Hendrickson framework correlates the baseline biomarker (e.g., a blood pressure summary) with the BR factor only, and treats the biomarker as conditionally independent of PB and TV. The justification is that the biomarker is intended to be a *pharmacological* predictor: it measures something about the participant's likelihood of responding to the drug's mechanism, not their suggestibility or their natural-history trajectory.

If a candidate biomarker turned out to be correlated with PB, that would be a substantive finding (the biomarker is partly psychological in its action), and the model could be re-specified to allow that correlation. The default specification reflects the prior that a useful predictive biomarker should be a clean pharmacological predictor; deviations from that prior are testable.

We must add a qualification which is not a matter of modeling preference. The implementation supplies an optional biomarker to placebo-belief correlation, and paper 06 exercises it. It supplies no biomarker to natural-history counterpart at all. The independence of biomarker and PB is therefore an assumption that can be relaxed, whereas the independence of biomarker and TV is at present a property of the software. Section 18.1 of Part V sets out why the difference matters.

Q: What if TV is negative (the patient naturally worsens)?

In the standard parameterization with positive m_{TV}, the Gompertz form is monotonically increasing and TV is non-negative by construction. To allow naturally-worsening trajectories, the maximum m_{TV} can be set negative (or, in an extended DGP, drawn from a population distribution that includes negative values; the current implementation uses a fixed value); the resulting Gompertz curve is then monotonically decreasing in symptom-reduction (equivalently, monotonically increasing in symptom severity). The model handles this case without modification, and detecting $\hat{m}_{TV} < 0$ is itself informative: it implies the underlying condition is deteriorating, and the apparent treatment effect is partly camouflaging deterioration that would otherwise be visible.

Q: How many timepoints do I need to identify all three components?

Roughly, at least one timepoint per phase contrast that you want to exploit. A pure open-label design with eight weekly timepoints can identify $BR + PB$ jointly but cannot separate them. Adding a blinded discontinuation phase with at least three or four timepoints lets the model identify BR separately. Adding a longer off-drug follow-up extends identifiability of TV, particularly if the participant's natural-history trajectory is still evolving. We note separately that

adding occasions tightens the positive-definiteness bound on the moderation parameter, and section 20 gives the closed form. Identifiability and feasibility therefore pull in opposite directions as the schedule grows. The design contrasts table of section 5 is the formal version of the answer: each component needs at least one phase in which it is uniquely present (or uniquely absent in a comparable arm), and the more such phases the better.

Q: What if my drug has a fast onset and a long washout?

The Gompertz parameters absorb this asymmetry. A drug with a fast BR onset has a high r; a drug with a long pharmacological washout has a long off-drug tail captured by the carryover half-life $t_{1/2}$ used in the analysis-side drug indicator $D_{bc,it} = \exp(-\lambda \cdot t_{sd})$ (off drug). The decomposition is agnostic to the specific pharmacokinetics so long as the trial includes enough timepoints in the relevant transition windows (on-to-off and off-to-on) to constrain the curves.

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24. Why this matters for biomarker validation

The motivating clinical question for the trial designs in this project is: 'does the biomarker predict treatment response?', and specifically 'does the biomarker predict the *pharmacological* component of treatment response?'. This is the precision-medicine question, and it depends critically on the BR-PB-TV decomposition.

The wrong way

A direct regression of the total response R_i on the baseline biomarker B_i,

$$R_i \;=\; \beta_0 + \beta_{bm}^{\text{lumped}}\, B_i + \xi_i,$$

estimates the biomarker's association with whatever the lumped total contains. If PB varies systematically across the biomarker distribution (because, say, the biomarker is correlated with optimism, suggestibility, or treatment-seeking history), then $\hat{\beta}_{bm}^{\text{lumped}}$ contains a placebo-prediction component that is inseparable from the pharmacological-prediction component. The resulting biomarker, applied prospectively, would mis-stratify patients: those flagged as 'high responders' by the biomarker would include both true high-BR responders and high-PB responders, and the latter would not benefit from prescription decisions made on a pharmacological basis.

The right way

Regress *each component* on the biomarker:

$$\begin{aligned} BR_i &= \beta_0^{BR} + \beta_{bm}^{BR}\, B_i + \xi_i^{BR}, \\ PB_i &= \beta_0^{PB} + \beta_{bm}^{PB}\, B_i + \xi_i^{PB}, \\ TV_i &= \beta_0^{TV} + \beta_{bm}^{TV}\, B_i + \xi_i^{TV}. \end{aligned}$$

(The symbol β_{bm} without a superscript is reserved for the mean-moderation multiplier, so the slopes here always carry one.)

A biomarker is a useful pharmacological predictor if and only if $\hat{\beta}_{bm}^{BR}$ is meaningfully non-zero. Significant $\hat{\beta}_{bm}^{PB}$ or $\hat{\beta}_{bm}^{TV}$ are diagnostic findings about the biomarker (it is partly psychological, or it correlates with natural-history characteristics), but they are not the validation of pharmacological prediction. A biomarker that correlates only with PB would be valuable for selecting placebo responders in run-in designs but worthless for predicting pharmacological efficacy; a biomarker that correlates only with TV would be a prognostic marker, not a predictive one.

The decomposition is therefore not optional for the precision- medicine question: it is the formal mechanism by which the question is asked at all.

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25. Summary table

Table 13: Summary of the BR, PB, TV and noise components

Component	Cause	Timescale	Identified by	Inferential role
BR	Drug pharmacology	Pharmacokinetic (hours-weeks)	Phase contrasts (on-drug vs. blinded placebo)	Pharmacological efficacy, biomarker-prediction target
PB	Belief in treatment	Expectancy-modulated (hours-weeks)	Open-label vs. blinded contrasts	Placebo response, accounted for, not the target

Component	Cause	Timescale	Identified by	Inferential role
TV	Natural history	Disease-trajectory (weeks-months)	Off-drug timepoints	Confounding control
ε	Measurement noise	Per-observation	Residuals	Fit diagnostics

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Key takeaway

The BR-PB-TV decomposition is not technical bookkeeping. It is the mechanism by which the trial design and the model together identify clinically distinct causes of symptom change from a single within-person trajectory. Without the decomposition, the analyst has only the lumped total, and the lumped total mixes pharmacological efficacy with placebo response and natural history. Pharmacological efficacy is the regulatory and clinical quantity of interest; the other two are biases that must be controlled for before efficacy can be inferred.

The trial design (open-label phase, blinded discontinuation, crossover) supplies the identifiability. The model extracts the components from the design contrasts. The biomarker validation question, if asked at the level of total response rather than the BR component, is unanswerable.

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26. References

Metadata of the published references below were verified on 2026-10-07 against PubMed or Crossref.

- Hendrickson, R. C., Thomas, R. G., Schork, N. J. and Raskind, M. A. (2020). Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: statistical methods and theoretical findings. *Frontiers in Digital Health* 2:13. doi:10.3389/fdgth.2020.00013. Source of the prazosin-PTSD Gompertz calibration used throughout this document. (Earlier versions gave the title as "Optimizing aggregated N-of-1 trial designs for the detection of biomarker-treatment interactions", which is not the published title.)

- Hróbjartsson, A. and Gøtzsche, P. C. (2001). Is the placebo powerless? An analysis of clinical trials comparing placebo with no treatment. *New England Journal of Medicine* 344:1594-1602. doi:10.1056/NEJM200105243442106.
- Hróbjartsson, A. and Gøtzsche, P. C. (2004). Is the placebo powerless? Update of a systematic review with 52 new randomized trials comparing placebo with no treatment. *Journal of Internal Medicine* 256:91-100. doi:10.1111/j.1365-2796.2004.01355.x.
- Hróbjartsson, A. and Gøtzsche, P. C. (2010). Placebo interventions for all clinical conditions. *Cochrane Database of Systematic Reviews* 2010(1):CD003974. doi:10.1002/14651858.CD003974.pub3. Outcome data from 202 of 234 included trials in 60 clinical conditions; the principal source for section 14.
- Kirsch, I., Deacon, B. J., Huedo-Medina, T. B., Scoboria, A., Moore, T. J. and Johnson, B. T. (2008). Initial severity and antidepressant benefits: a meta-analysis of data submitted to the Food and Drug Administration. *PLoS Medicine* 5(2):e45. doi:10.1371/journal.pmed.0050045.
- Locher, C., Kossowsky, J., Gaab, J., Kirsch, I., Bain, P. and Krummenacher, P. (2015). Moderation of antidepressant and placebo outcomes by baseline severity in late-life depression: a systematic review and meta-analysis. *Journal of Affective Disorders* 181:50-60. doi:10.1016/j.jad.2015.03.062. With Kirsch et al. (2008), antidepressant trial analyses bearing on the separation of placebo response from regression to the mean.
- Senn, S. (2002). *Cross-over Trials in Clinical Research*, 2nd ed. Chichester: Wiley. Principal source for section 15. (Book; metadata not checked against Crossref.)
- Senn, S. (2016). Mastering variation: variance components and personalised medicine. *Statistics in Medicine* 35:966-977. doi:10.1002/sim.6739.
- Schmid, C. H. and Yang, J. (2022). Bayesian models for N-of-1 trials. *Harvard Data Science Review*, Special Issue 3. doi:10.1162/99608f92.3f1772ce. Cited in section 15 for an individual-level trend parameter alongside treatment effect and autocorrelation (that content claim was not re-checked against the article).
- Colloca, L., Lopiano, L., Lanotte, M. and Benedetti, F. (2004). Overt versus covert treatment for pain, anxiety, and Parkinson's disease. *Lancet Neurology* 3:679-684. doi:10.1016/S1474-4422(04)00908-1.
- Rohsenow, D. J. and Marlatt, G. A. (1981). The balanced placebo design: methodological considerations. *Addictive Behaviors* 6:107-122. doi:10.1016/0306-4603(81)90003-4. With Colloca et al. (2004), the open-hidden and balanced-placebo designs cited in section 7 as the established means of separating drug exposure from belief. (Earlier versions listed Rohsenow as sole author.)
- pmsimstats team. Paper 06, Three-component decomposition of treatment response in aggregated N-of-1 trials (`analysis/report/06-component-decomposition/`). Supplies the identifiability conditions, the omitted-variable-bias identity,

Study A, the contamination pilot and Study B.

- pmsimstats team. Paper 13, dropping the placebo-belief component (`analysis/report/13-two-component-drop-pb/`), and Paper 14, dropping natural history (`analysis/report/14-two-component-drop-tv/`). The evidence base for Part V.
- pmsimstats team. Component-decomposition pedagogy (`docs/24-component-decomposition-pedagogy`). The standalone introduction from which Parts I through III are drawn.